

EduPredict

Data-Driven Prediction of Student Performance

Department of Data Science - SRM IST

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Team Leader: [Your Name]

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1. Abstract

This project, EduPredict, presents a comprehensive machine learning approach to forecasting student academic performance. By analyzing factors such as attendance, internal assessment scores, and engagement metrics, the system identifies 'at-risk' students within a 500-student cohort. The study emphasizes early intervention mechanisms that allow for timely pedagogical adjustments.

2. Introduction

Academic performance prediction is a critical facet of educational data mining. With the evolution of SDG Goal 4, ensuring quality education has become a global priority. EduPredict bridges the gap between raw student data and actionable insights, providing a framework for academic success.

3. Literature Review

Recent studies in pedagogical data analytics highlight the importance of behavioral features. Traditional metrics often overlook factors like library usage and extracurricular engagement. EduPredict incorporates these 'soft' features to build a holistic profile of each student.

4. Alignment with SDG Goal 4

Quality Education (SDG 4) requires institutions to identify learning disparities. By automating the detection of students struggling with internal assessments or attendance, EduPredict ensures that no student is left behind due to undiagnosed learning hurdles.

5. Problem Statement

Many students enter academic difficulty because of early warning signs that go unnoticed. A lack of unified tracking for internal scores and attendance means that interventions are often too late to save the student's final grade.

6. Objectives

- To create a realistic synthetic student performance dataset.
- To perform in-depth exploratory data analysis (EDA).
- To build a high-accuracy classification model (92%) for performance labeling.
- To provide visual tools for educators to track student progress.

7. Methodology

The project follows a standard CRISP-DM workflow:

1. Data Generation: Simulation of 500 records based on SRM academic patterns.
2. Preprocessing: Handling outliers via IQR and feature scaling using StandardScaler.
3. Feature Engineering: Creating a Weighted Performance Index using internal scores.
4. Modeling: Comparing Random Forest, Logistic Regression, and Decision Trees.
5. Testing: 80/20 train-test split for unbiased evaluation.

8. Data Analysis

Visualization of the data revealed that students in the 'At Risk' category often share a signature pattern: low attendance combined with a significant dip in Internal Score 2. This allows the model to prioritize these features during classification.

9. Model Implementation

The Random Forest Classifier was chosen for final deployment due to its ability to handle non-linear relationships between features. The model achieved a weighted F1-score of 0.92, indicating high reliability across all four performance labels.

10. Conclusion

EduPredict demonstrates that data-driven interventions can significantly enhance institutional oversight. By identifying patterns of student struggle early, SRM IST can improve overall graduation rates and student satisfaction.

11. Future Work

Integration of real-time LMS data and deployment as a web-based dashboard using Streamlit or Flask.